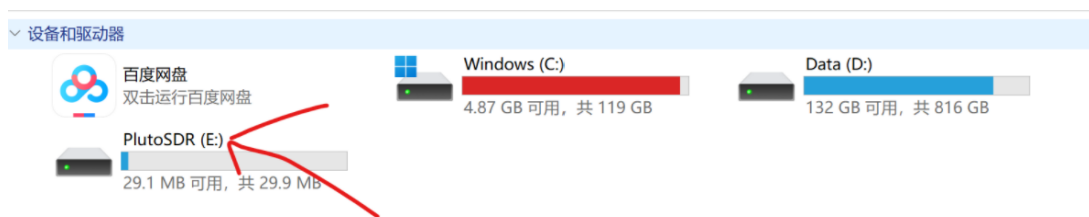


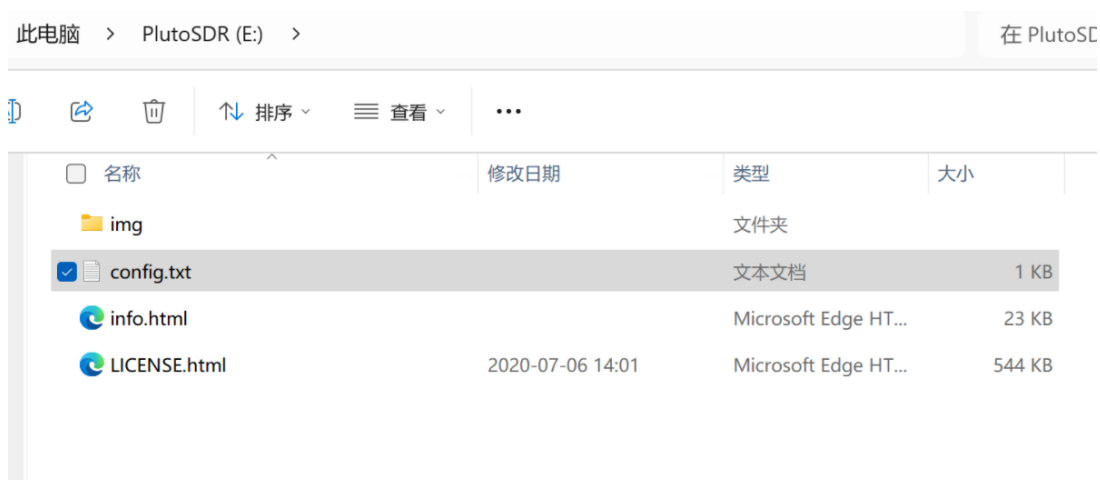
先连接第一个设备，见下图：

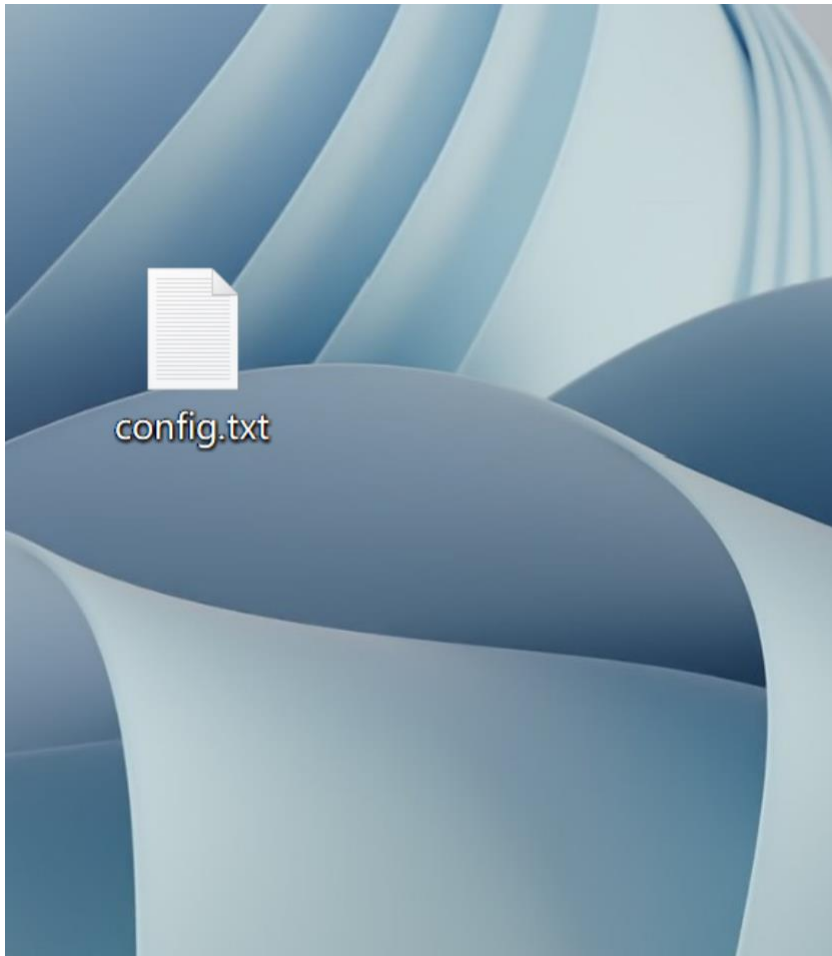


双击打开，读取到的 Pluto 设备，见下图：



然后把 config.txt 文件拷贝到桌面，见下图：





拷贝过后双击打开桌面的 config.txt 文件，将 ipaddr 设成 192.168.3.2（作为接收），然后保存，见下图：

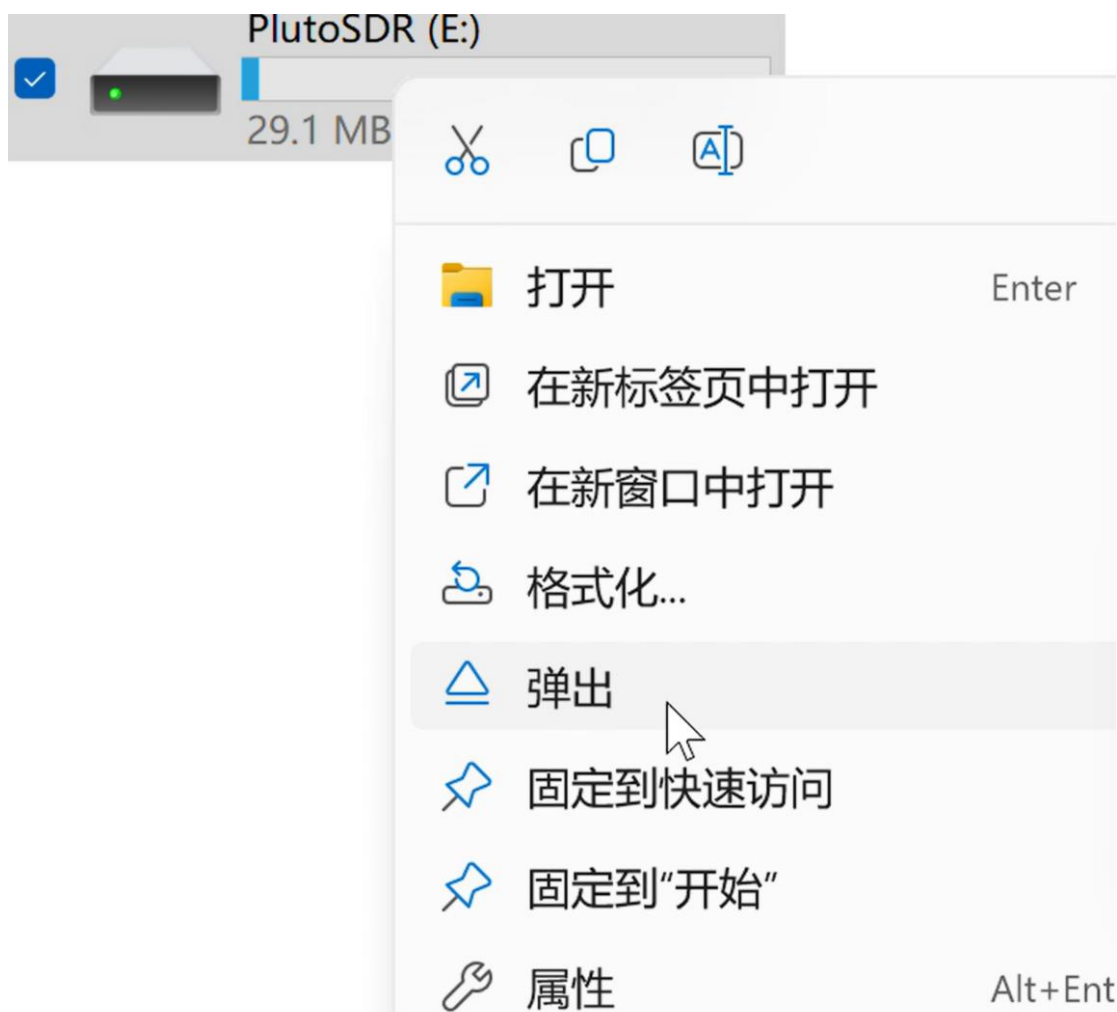
```
# Analog Devices PlutoSDR Rev.C (Z7010-AD9363)
# Device Configuration File
# 1. Open with an Editor
# 2. Edit this file
# 3. Save this file on the device USB drive
# 4. Eject the device USB Drive
# Doc: https://wiki.analog.com/university/tools/pluto/users/customizing

[NETWORK]
hostname = pluto
ipaddr = 192.168.3.2
ipaddr_host = 192.168.3.10
netmask = 255.255.255.0
```

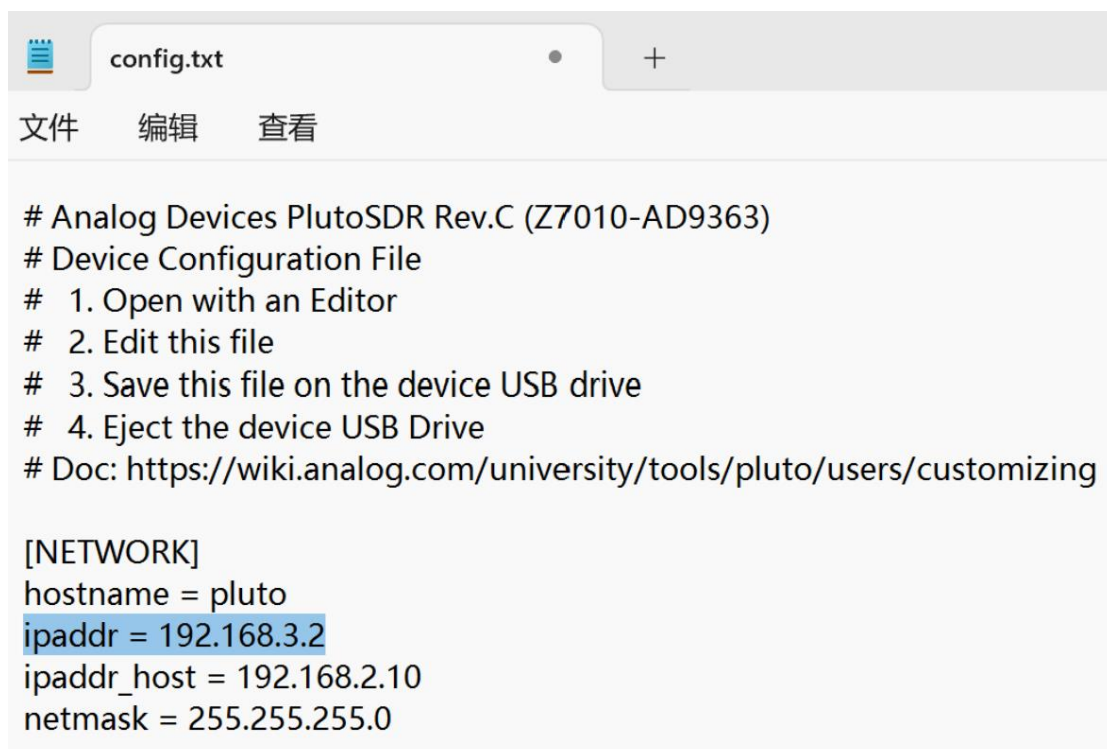
然后将桌面的 config.txt 文件替换之前 config.txt 文件，见下图：



弹出刚才修改 config 的设备（注意，要用弹出的，最好不要直接拔），见下图：



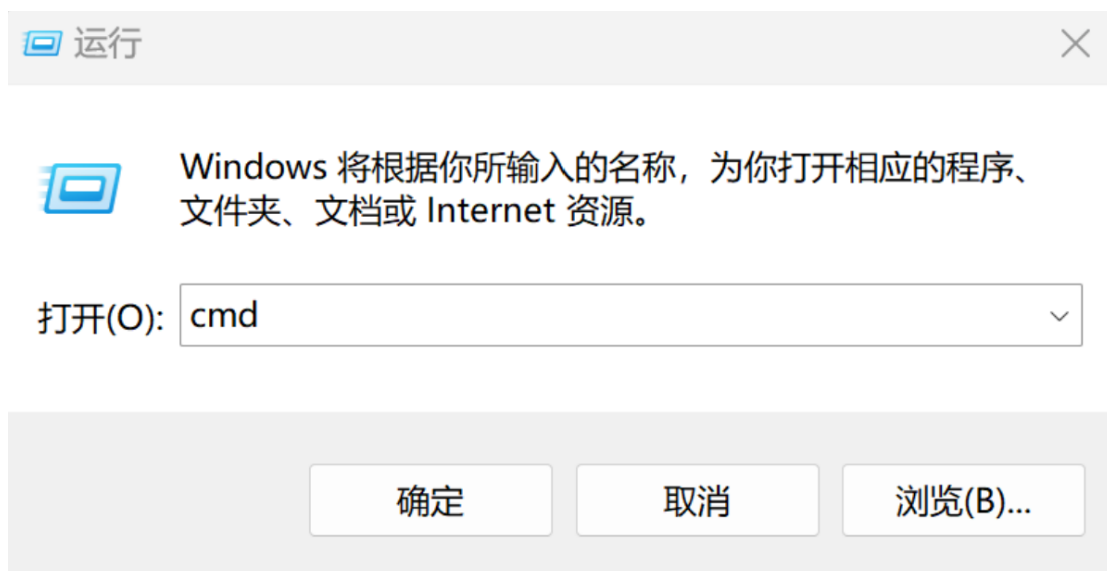
弹出后再重新插上，查看刚才的 ip 是否修改成功，可见 ip 仍然是 192.168.3.2，说明修改成功，见下图：



```
# Analog Devices PlutoSDR Rev.C (Z7010-AD9363)
# Device Configuration File
# 1. Open with an Editor
# 2. Edit this file
# 3. Save this file on the device USB drive
# 4. Eject the device USB Drive
# Doc: https://wiki.analog.com/university/tools/pluto/users/customizing

[NETWORK]
hostname = pluto
ipaddr = 192.168.3.2
ipaddr_host = 192.168.2.10
netmask = 255.255.255.0
```

打开 cmd，见下图：



运行

Windows 将根据你所输入的名称，为你打开相应的程序、文件夹、文档或 Internet 资源。

打开(O):

确定 取消 浏览(B)...

输入 conda activate base，见下图：

```
C:\WINDOWS\system32\cmd. X + v

Microsoft Windows [版本 10.0.22631.3737]
(c) Microsoft Corporation。保留所有权利。

C:\Users\wyg_2>conda activate base

(base) C:\Users\wyg_2>
```

输入 python，见下图：

```
(base) C:\Users\wyg_2>python
Python 3.7.9 (default, Aug 31 2020, 17:10:11) [MSC v.1916 64 bit (AMD64)] :: Anaconda, Inc. on win32
Type "help", "copyright", "credits" or "license" for more information.
>>>
```

输入 import adi，见下图：

```
>>> import adi
>>>
```

再输入 sdr = adi.Pluto('ip:192.168.3.2')，没有异常出现，说明设备 ip 修改成功，并记住这台设备是接收设备，见下图：

```
>>> sdr = adi.Pluto('ip:192.168.3.2')
>>> |
```

插入另一台设备，见下图：





用同样的方法把另一台设备的 ip 设置为 192.168.2.1，见下图：

```
config.txt
文件 编辑 查看

# Analog Devices PlutoSDR Rev.C (Z7010-AD9363)
# Device Configuration File
# 1. Open with an Editor
# 2. Edit this file
# 3. Save this file on the device USB drive
# 4. Eject the device USB Drive
# Doc: https://wiki.analog.com/university/tools/pluto/users/customizing

[NETWORK]
hostname = pluto
ipaddr = 192.168.2.1
ipaddr_host = 192.168.2.10
netmask = 255.255.255.0
```

再输入 `sdr = adi.Pluto('ip:192.168.2.1')`，可见发射设备也能搜索得到，见下图：

```
>>> sdr = adi.Pluto('ip:192.168.2.1')
>>>
```

最后先后输入 `sdr = adi.Pluto('ip:192.168.2.1')`和 `sdr = adi.Pluto('ip:192.168.3.2')`，发现均

能通信，说明两台设备连接正常，见下图：

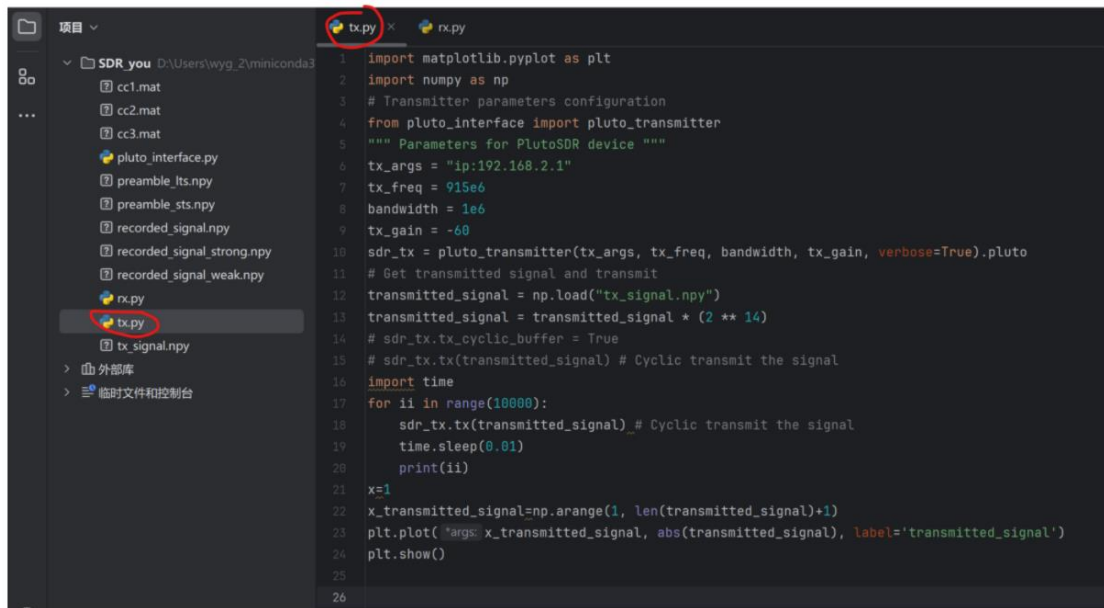
```
>>> sdr = adi.Pluto('ip:192.168.2.1')
>>> sdr = adi.Pluto('ip:192.168.3.2')
>>>
```

然后分清楚发射和接收设备，把发射设备的发射端通过线缆直接连接到接收设备的接收端，见下图：



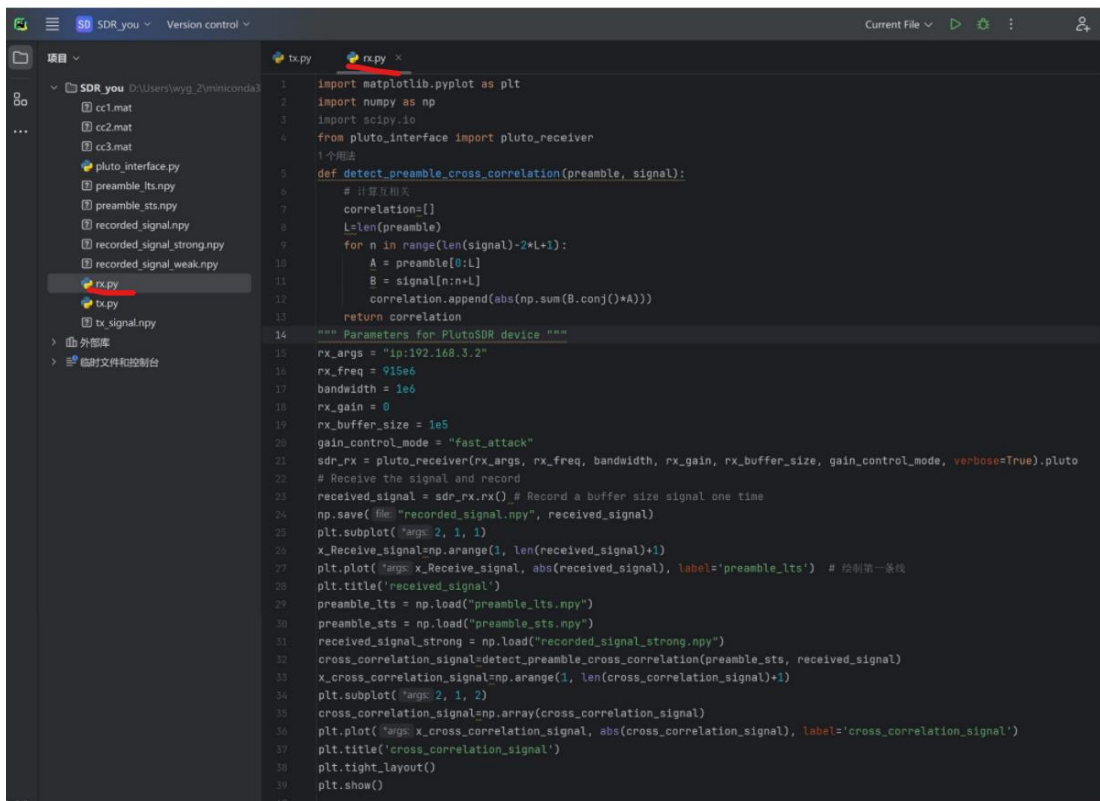


把发射部分的代码拷贝到 pycharm 中的 tx.py 文件内，见下图：

A screenshot of the PyCharm IDE. The left sidebar shows a project named 'SDR\_you' with various files. The file 'tx.py' is selected and highlighted with a red circle. The main editor window displays the code for tx.py, which includes imports for matplotlib and numpy, configuration for a PlutoSDR transmitter, and a loop for transmitting a signal. The code is as follows:

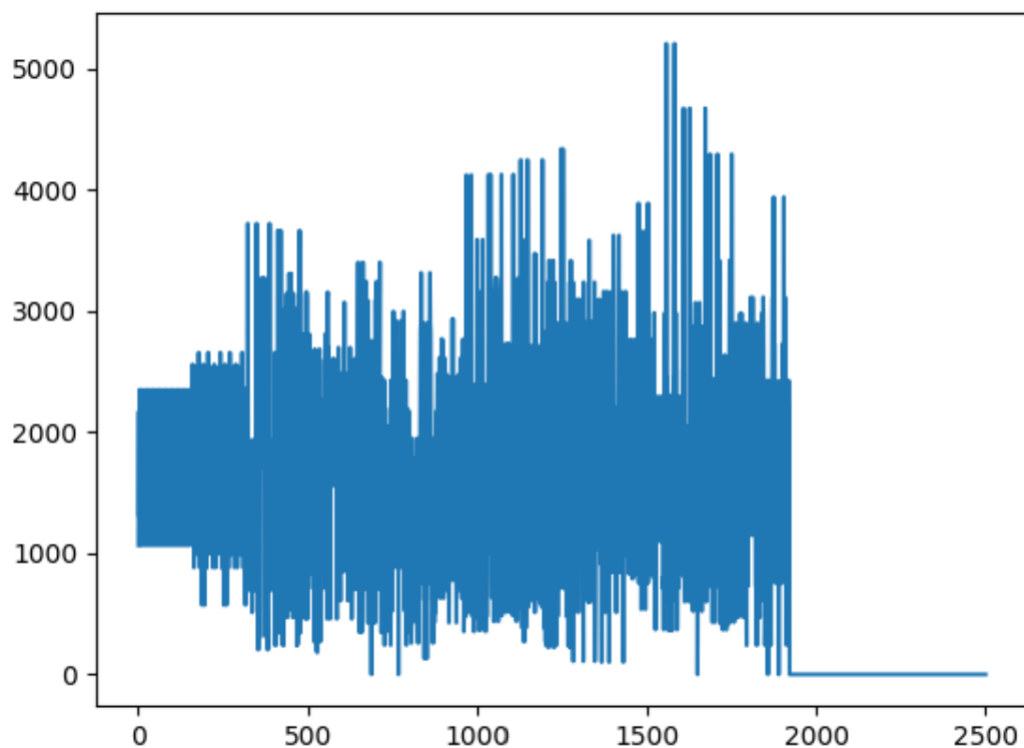
```
1 import matplotlib.pyplot as plt
2 import numpy as np
3 # Transmitter parameters configuration
4 from pluto_interface import pluto_transmitter
5 """ Parameters for PlutoSDR device """
6 tx_args = "ip:192.168.2.1"
7 tx_freq = 915e6
8 bandwidth = 1e6
9 tx_gain = -60
10 sdr_tx = pluto_transmitter(tx_args, tx_freq, bandwidth, tx_gain, verbose=True).pluto
11 # Get transmitted signal and transmit
12 transmitted_signal = np.load("tx_signal.npy")
13 transmitted_signal = transmitted_signal * (2 ** 14)
14 # sdr_tx.tx_cyclic_buffer = True
15 # sdr_tx.tx(transmitted_signal) # Cyclic transmit the signal
16 import time
17 for ii in range(10000):
18     sdr_tx.tx(transmitted_signal) # Cyclic transmit the signal
19     time.sleep(0.01)
20     print(ii)
21 x=1
22 x_transmitted_signal=np.arange(1, len(transmitted_signal)+1)
23 plt.plot(x_transmitted_signal, abs(transmitted_signal), label='transmitted_signal')
24 plt.show()
25
26
```

把接收部分的代码拷贝到 pycharm 中的 rx.py 文件内，见下图：

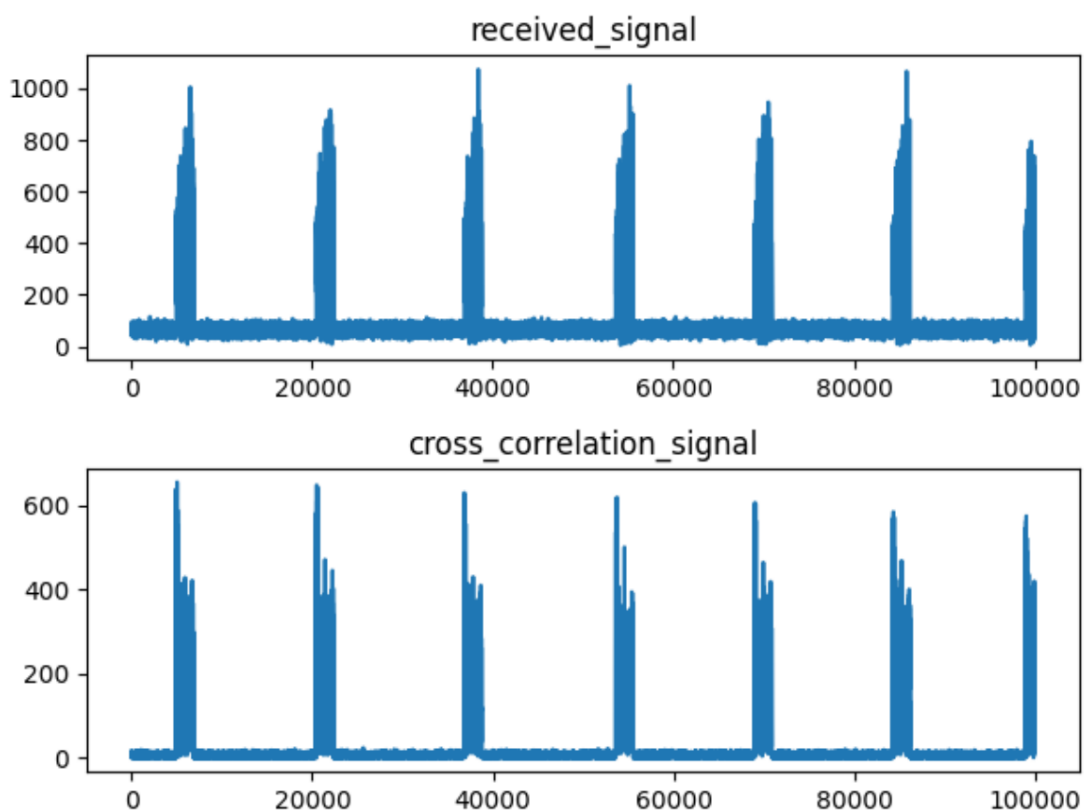
A screenshot of the PyCharm IDE. The left sidebar shows the same project 'SDR\_you'. The file 'rx.py' is selected and highlighted with a red circle. The main editor window displays the code for rx.py, which includes imports for matplotlib, numpy, and scipy, a function to detect preamble cross-correlation, and configuration for a PlutoSDR receiver. The code is as follows:

```
1 import matplotlib.pyplot as plt
2 import numpy as np
3 import scipy.io
4 from pluto_interface import pluto_receiver
5 # 1个用途
6 def detect_preamble_cross_correlation(preamble, signal):
7     # 计算互相关
8     correlations=[]
9     L=len(preamble)
10    for n in range(len(signal)-2*L+1):
11        A = preamble[0:L]
12        B = signal[n:n+L]
13        correlation.append(abs(np.sum(B.conj()*A)))
14    return correlation
15 """ Parameters for PlutoSDR device """
16 rx_args = "ip:192.168.3.2"
17 rx_freq = 915e6
18 bandwidth = 1e6
19 rx_gain = 0
20 rx_buffer_size = 1e5
21 gain_control_mode = "fast_attack"
22 sdr_rx = pluto_receiver(rx_args, rx_freq, bandwidth, rx_gain, rx_buffer_size, gain_control_mode, verbose=True).pluto
23 # Receive the signal and record
24 received_signal = sdr_rx.rx() # Record a buffer size signal one time
25 np.save("recorded_signal.npy", received_signal)
26 plt.subplot("args: 2, 1, 1)
27 x_receive_signal=np.arange(1, len(received_signal)+1)
28 plt.plot(x_receive_signal, abs(received_signal), label='preamble_lts') # 绘制第一条线
29 plt.title('received_signal')
30 preamble_lts = np.load("preamble_lts.npy")
31 preamble_sts = np.load("preamble_sts.npy")
32 received_signal_strong = np.load("recorded_signal_strong.npy")
33 cross_correlation_signal=detect_preamble_cross_correlation(preamble_sts, received_signal)
34 x_cross_correlation_signal=np.arange(1, len(cross_correlation_signal)+1)
35 plt.subplot("args: 2, 1, 2)
36 cross_correlation_signal=np.array(cross_correlation_signal)
37 plt.plot(x_cross_correlation_signal, abs(cross_correlation_signal), label='cross_correlation_signal')
38 plt.title('cross_correlation_signal')
39 plt.tight_layout()
40 plt.show()
41
```

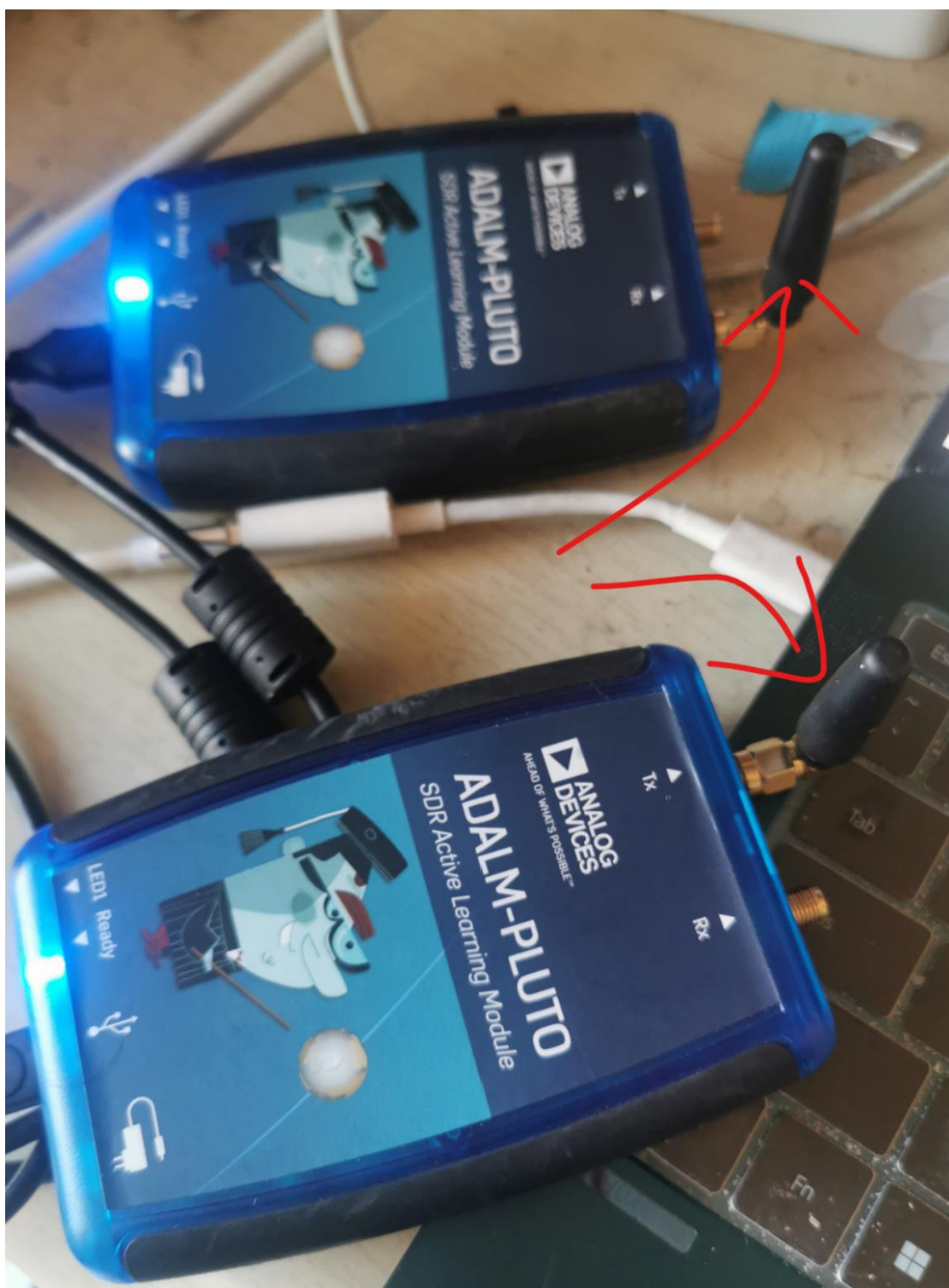
先运行 tx.py 部分代码，出现右图，见下图（为发送的发射信号）：



再运行 rx.py 部分代码，出现右图，右上图是接收到的发射器发射的信号，右下图是做完自相关以后的信号，见下图：



再把直连的线缆换成无线发射，见下图：



得到接收图，可见多了噪声，见下图：

